



# ANGELA KATOLA WANJALA

## Data Analyst

### Contacts

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[Github](#)

### Education

#### AkiraChix

codeHive- Diploma in Information Technology  
February 2025-Present

### Skills

#### Languages:

Python, SQL

#### Libraries & Frameworks:

Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, HuggingFace Transformers, Matplotlib, Seaborn, Django, Next.js, React.js

#### Databases & Vector Search:

PostgreSQL, ChromaDB

#### Platforms & Tools:

Jupyter Notebook/Lab, Google Colab, GitHub, Postman, figma

### Referees

Linda Kamau

Founder and Executive Director

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Caroline Mutua

Product Manager

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### Personal Statement

In Africa, agriculture is more than a livelihood; it is the backbone of economies and the key to unlocking sustainable development. For Angela, this isn't just a fact but an opportunity: if agriculture sustains the continent, then technology is what drives it forward. She is motivated to use her skills to contribute to a sector uniquely positioned to exploit new technologies and innovations. Her excitement lies at this turning point where tech becomes the game-changer for agriculture.

### Projects

#### Recos AI

Building Recos, an AI-powered agent designed to assist recruiters in identifying the ideal candidates through objective, data-driven insights that reduce skill mismatches and enhance hiring precision.

#### Approach & Key Learnings:

- Integrated Odoo recruitment data with a PostgreSQL backend, implementing an automated, real-time data synchronization system that ensures accurate and up-to-date candidate information for seamless analysis.
- Leveraged the Google Calendar API for automated interview scheduling and developed a Google Meet add-on featuring live transcription and real-time interview analytics, significantly improving the interview experience and operational efficiency.
- Developed an NLP system using Gemini LLM to accurately extract candidate skills from unstructured resumes and perform meaningful matching, enabling precise identification of competencies and generation of relevant interview questions in real time.
- Optimized prompt engineering with Gemini LLM for consistent skill extraction accuracy and reliable sentiment classification across different candidate profiles and interview scenarios.

#### Movie Recommendation System (RAG + AI Agent)

Built a personalized movie recommendation system that suggests films based on user queries by leveraging semantic search and AI-generated explanations to enhance relevance and user engagement.

#### Approach & Key Learnings:

- Used SentenceTransformer (all-MiniLM-L6-v2) to embed movie title and description chunks and stored embeddings in ChromaDB for fast, scalable vector search.
- Implemented efficient batch embedding and indexing to handle large datasets while maintaining performance.
- Learnt text preprocessing techniques including cleaning, lemmatization, and chunking to optimize embedding quality and retrieval relevance.
- Tested prompting strategies and used zero-shot prompting with clear, detailed instructions to guide the model. This approach proved effective for generating creative, context-aware movie explanations without requiring example demonstrations.

#### Housing Prices Prediction

Developed a machine learning pipeline to predict housing prices by preparing and examining data from various sources, revealing significant patterns.

#### Approach & Key Learnings:

- Cleaned and preprocessed housing datasets using Pandas, addressing missing values with median and mode imputations to ensure accuracy, reliability and bias prevention.
- Created visualizations using Matplotlib and Seaborn to analyze SalePrice distributions and the breakdown of different neighborhoods, enabling data-driven insights for model improvement.
- Performed correlation analysis through heatmaps to identify and remove features that were too similar to each other, which helped make the model more stable and reliable.
- Trained and evaluated multiple regression models such as Linear Regression, Random forest and KNN Regression models using scikit-learn, for precise house price predictions on the test set.
- Assessed model performance with  $R^2$  and RMSE metrics, achieving reliable house price predictions through iterative model refinement.